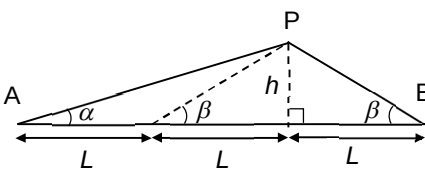
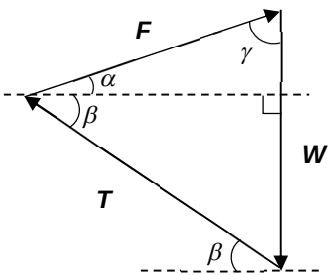


Qns	Answer	Marks
1(a)	Resultant force acting on the body is zero. Resultant moment OR Torque <u>about any point</u> is zero.	B1 B1
1(b)	Point at which <u>whole weight of body may be considered / seems to act.</u>	B1
1(c)(i)	Since T and W have zero moment about P (because their line of actions passes through P), so F must have zero moment, i.e. line of action of F must pass through P OR When all forces have lines of actions that pass through P, (perpendicular) distance from P is zero for all forces (M1), so sum of moments about P is zero	B1 B1

1(c)(ii)1.	By principle of moment and taking moments about A, $2W = 3T \sin \beta$ $2(10) = 3T \sin 30^\circ$ $T = 13.3 \text{ N}$	B1 A1
1(c)(ii)2.	Sum of forces in the vertical direction and horizontal direction are both zero. Considering horizontal forces: $F \cos \alpha = T \cos \beta \quad (1)$ Considering vertical forces: $W = F \sin \alpha + T \sin \beta \Rightarrow F \sin \alpha = W - T \sin \beta \quad (2)$ $\frac{\text{Eq(2)}}{\text{Eq(1)}}: \tan \alpha = \frac{W - T \sin \beta}{T \cos \beta} = \frac{10 - 13.3 \sin 30}{13.3 \cos 30}$ $\alpha = 16.2^\circ$ (Physics statements to be present to score full credit) <u>Alternative method (for both parts):</u>  $\tan \alpha = \frac{h}{2L}$ $\tan \beta = \frac{h}{L}$ $\tan \alpha = 0.5 \tan \beta$ $= 0.5 \tan 30^\circ$ $\alpha = 16^\circ$  $\gamma = 90^\circ - \alpha = 90^\circ - 16^\circ = 74^\circ$ Using sine rule, $\frac{T}{\sin \gamma} = \frac{W}{\sin(\alpha + \beta)}$ $T = \frac{10}{\sin(30^\circ + 16^\circ)} \times \sin 74^\circ$ $= 13.4 \text{ N}$	B1 Horizontal B1 Vertical A1

Qns	Answer	Marks
2(a)(i)	An arrow originating from the object pointing towards point O and	B1